

HOT SPRINGS, AR, USA

WMO: 723415

Lat: 34.478N

Lon: 93.096W

Elev: 163

StdP: 99.38

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 03962

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-3.8	-14.8	1.1	-2.3	-12.2	1.3	-0.2	8.0	9.5	7.3	9.5	1.8	80	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	37.0	24.1	35.6	24.3	34.0	24.1	26.3	32.4	25.8	31.9	25.3	31.4	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.1	28.3	24.1	19.4	28.2	23.7	18.9	28.0	82.1	32.6	80.2	32.0	78.2	31.3	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.0	5.4	-9.6	38.9	2.0	2.2	-11.0	40.5	-12.2	41.9	-13.3	43.1	-14.8	44.8		
(5)			-10.6	27.1	1.8	0.7	-11.9	27.6	-13.0	28.0	-14.0	28.4	-15.3	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	5.9	7.6	12.6	17.4	21.7	26.3	28.1	27.8	24.2	17.8	11.8	7.4		
	DBStd	8.99	5.44	5.55	5.18	4.24	3.69	2.44	2.44	2.61	3.34	4.60	4.95	4.92		
	HDD10.0	442	151	104	32	3	1	0	0	0	0	2	37	113		
	HDD18.3	1573	387	303	189	66	15	0	0	0	2	68	203	341		
	CDD10.0	3151	23	37	112	225	362	491	561	552	425	243	90	31		
	CDD18.3	1241	1	1	11	38	118	241	303	294	177	51	7	1		
	CDH23.3	12934	1	7	68	307	979	2549	3502	3347	1707	435	32	1		
	CDH26.7	5840	0	0	6	50	298	1164	1769	1689	731	130	2	0		
(14) Wind	WSAvg	2.3	2.4	2.6	2.8	2.7	2.3	2.1	1.9	1.9	2.0	2.1	2.2	2.3		
(15) Precipitation	PrecAvg	1391	104	95	135	136	161	103	112	72	101	126	125	126		
	PrecMax	2088	240	203	262	268	395	258	291	221	285	374	296	257		
	PrecMin	942	14	28	35	56	15	34	4	13	11	10	35	15		
	PrecStd	269	55	39	62	66	86	58	59	47	60	84	68	57		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	24.2	27.5	29.8	32.6	37.1	38.9	39.0	36.4	32.3	26.8	22.6		
		MCWB	16.6	17.0	17.9	19.9	22.3	23.5	23.9	24.6	23.2	21.0	19.4	17.6		
	2%	DB	19.9	21.5	25.2	27.9	31.2	35.0	36.9	37.2	34.3	30.0	23.8	20.2		
		MCWB	15.2	16.1	17.0	19.1	21.9	24.0	24.2	24.3	23.0	21.4	17.6	16.8		
	5%	DB	17.8	19.1	23.0	26.9	29.8	33.4	35.3	35.6	32.6	27.8	22.2	18.0		
		MCWB	14.4	14.8	15.9	18.6	21.8	23.7	24.6	24.5	23.0	20.0	17.4	14.7		
	10%	DB	15.7	17.3	21.3	25.1	28.1	32.3	33.8	33.8	31.1	25.9	20.1	16.1		
		MCWB	10.8	12.9	15.2	17.8	21.2	23.5	24.6	24.3	22.5	19.2	15.8	12.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	19.5	20.0	22.6	24.7	26.4	26.9	26.9	25.5	24.1	21.6	19.5		
		MCDB	20.8	21.6	25.1	27.2	29.2	32.5	33.3	32.6	31.2	28.9	24.4	21.3		
	2%	WB	17.3	17.9	18.8	21.2	23.9	25.6	26.3	26.3	24.8	23.1	19.8	17.8		
		MCDB	18.9	20.1	23.3	25.5	28.3	31.6	32.6	32.4	30.5	27.3	22.7	19.6		
	5%	WB	15.2	15.8	17.7	20.4	23.1	25.1	25.8	25.7	24.2	22.0	18.3	15.6		
		MCDB	17.2	18.4	21.4	24.6	27.6	30.9	31.9	31.9	29.5	25.8	20.9	17.5		
	10%	WB	11.5	13.7	16.5	19.4	22.3	24.4	25.4	25.2	23.7	20.8	16.6	13.2		
		MCDB	13.9	16.1	20.3	23.5	26.8	30.0	31.4	31.5	28.6	23.9	19.4	15.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	10.4	11.5	11.7	10.5	10.4	10.6	10.7	11.3	11.7	11.1	10.3		
		MCDBR	13.4	13.1	13.3	13.0	11.8	12.0	12.4	12.8	13.0	13.6	12.6	12.5		
		MCWBR	9.4	8.8	7.2	6.1	4.6	3.7	3.1	3.4	4.0	6.0	7.6	9.1		
	5% WB	MCDBR	10.1	10.6	10.8	9.9	9.6	10.0	10.5	10.8	10.3	10.4	9.8	10.4		
		MCWBR	9.4	8.7	7.1	5.8	4.7	3.8	3.5	3.6	3.7	5.7	7.8	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.327	0.356	0.391	0.423	0.460	0.478	0.476	0.405	0.357	0.334	0.312		
	taud		2.497	2.486	2.428	2.332	2.304	2.224	2.211	2.199	2.379	2.488	2.505	2.540		
	Ebn at Noon		885	922	917	895	867	830	812	808	856	876	865	878		
	Edh at Noon		84	95	109	126	131	142	143	142	113	94	83	77		
(44) All-Sky Solar Radiation	RadAvg		2.53	2.95	4.11	5.15	5.61	6.32	6.18	5.70	4.90	3.80	2.71	2.21		
	RadStd		0.23	0.42	0.34	0.32	0.49	0.55	0.37	0.43	0.40	0.54	0.32	0.26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	+0.90	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page